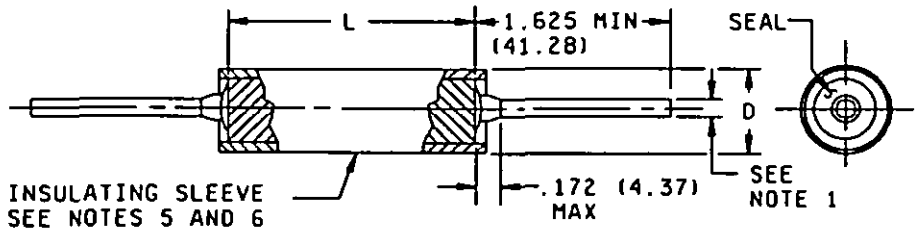
NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

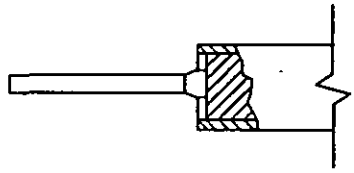
PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PAPER-PLASTIC FILM DIELECTRIC,
DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES),
ESTABLISHED RELIABILITY
STYLE CHR10 (INSULATED)

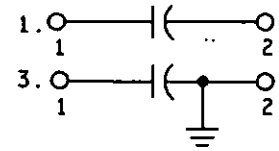
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39022.



CASE FOR CIRCUIT 1



CASE FOR CIRCUIT 3



CIRCUIT DIAGRAM

Figure 1: Dimensions and Configuration

NOTES:
1. Number 24 AWG wire for case diameters of .175 (4.45 mm) and .195 (4.95 mm).
   Number 22 AWG wire for case diameters of .235 (5.97 mm) and .312 (7.92 mm).
   Number 20 AWG wire for case diameters of .400 (10.16 mm) and over.
   (See Table I for exceptions.)
   Number 18 AWG wire: See Table I.
2. See Table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch = 25.4 mm.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 (.79 mm) inch on either end. Insulating sleeve thickness shall not exceed .016 (.41 mm) inch.
6. Plastic insulating sleeve shall be transparent; marking shall be placed on the capacitor case.
7. Metric equivalents are in parentheses

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Requirements:

Dimensions and configuration: See Figure 1 and Table I.
Case material: Nonmagnetic
Capacitance value See Table I.
Capacitance tolerance: 1 percent, 2 percent, 5 percent, and 10 percent
Rated voltage: See Table I for 85°C rating; derate to 60 percent of 85°C rating for operation at 125°C.
Dielectric material: Normally polycarbonate.
Operating temperature range: -55° to +125°C.

Failure rate level: M(1.0 percent), P(0.1 percent), R(0.01 percent), S(0.001 percent).

DC Burn-in: In accordance with MIL-C-39022, Table III, Qualification Inspection, Group I, and Table IV, Group A Inspection, Subgroup 1, except 140 percent of DC rated voltage shall be applied for 48 hours minimum at +125° +4, -0°C. 100 percent inspection required.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202; 100 percent inspection required following DC burn-in.

Terminal to Terminal:

AC: 100 ±10 Hz square wave, peak-to-peak voltage, three times DC rated voltage for 60 seconds minimum, but need not exceed 800 V peak-to-peak. Square wave signal shall have a rise time of 10 to 100 microseconds and a maximum overshoot of 20 percent of rated peak voltage.

DC: 200 percent of DC rated voltage for 60 seconds, minimum.

Terminals to case (when case is not a terminal): Same requirements as specified for terminal to terminal DC DWV Test.

Radiographic inspection (X-Ray): Method 209 of MIL-STD-202 (Required for FR levels R and S only).
Thermal shock: Method 107 of MIL-STD-202, Condition A, except Step 3 shall be +125° +4, -0°C.
Seal: Method 112 of MIL-STD-202, Condition A, except at +125° +4, -0°C, 1 hour with no evidence of leakage.
Insulation resistance (IR): Method 302 of MIL-STD-202.
Terminal to terminal - See Table III.
Terminal to case (when case is not a terminal) - 50,000 megohms, minimum.
Capacitance (CAP): Method 305 of MIL-STD-202.
Dissipation factor (DF):
At 25°C - 0.25 percent maximum
At 85°C - 0.40 percent maximum
At 125°C - 0.60 percent maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Table I. Capacitor characteristics and dimensions

DC Rated Voltage	Cap Value in µf	Dimensions, Nominal D +.015, -.005	L ±.031
50	.047	.174	.531
	.056	.174	.531
	.068	.174	.625
	.082	.174	.625
	.10	.174	.625
	.12	.174	.625
	.15	.193	.625
	.18	.193	.625
	.22	.235	.625
	.27	.235	.625
	.33	.312	.625
	.39	.312	.625
	.47	.312	.625
	.56	.312	.625
	.68	.312	.843
	.82	.312	.843
	1.0	.312	.843
	1.2	.400	.843
	1.5	.400	.843
	1.8	.400	.843
	2.0	.400	.843
	2.2	.400	1.125
	2.7	.400	1.125
	3.0	.400	1.125
	3.3	.400	1.125
	3.9	.500	1.125
	4.7	.500	1.125
	5.0	.500	1.125
	5.6	.500	1.125
	6.8	.562	1.125
	8.2	.562	1.312
	10.0	.670	1.312
	12.0	.670	1.312
	15.0	.750	1.375
	18.0	.750	1.375
	20.0	.750	1.625
	22.0	.750	1.625

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Table I. Capacitor characteristics and dimensions (cont'd)

DC Rated Voltage	Cap Value in µf	Dimensions, Nominal D +.015, -.005	L ±.031
100	.01	.174	.625
	.012	.174	.625
	.015	.174	.625
	.018	.174	.625
	.022	.174	.625
	.027	.174	.625
	.033	.174	.625
	.039	.174	.625
	.047	.193	.625
	.056	.193	.625
	.068	.235	.625
	.082	.235	.625
	.10	.235	.688
	.12	.235	.688
	.15	.312	.625
	.18	.312	.625
	.22	.312	.688
	.27	.312	.688
	.33	.312	.812
	.39	.312	.812
	.47	.400	.688
	.56	.400	.812
	.68	.400	.812
	.82	.400	.938
	1.0	.400	.938
	1.2	.500	.938
	1.5	.500	.938
	1.8	.500	1.125
	2.0	.500	1.125
	2.2	.500	1.125
	2.7	.562	1.312
	3.0	.562	1.312
	3.3	.562	1.312
	4.0	.562	1.562
	4.7	.670	1.312
	5.0	.670	1.312
	5.6	.670	1.312
	6.8	.670	1.562
	8.2	.670	1.812
	10.0	.750	1.812
	12.0	.750	1.812
	15.0	1.000	1.875
	18.0	1.000	1.875
	20.0	1.000	1.875
	22.0	1.000	1.875

Case size of 1" dia uses #18 AWG leads.

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Table I. Capacitor characteristics and dimensions (cont'd)

DC Rated Voltage	Cap Value in µf	Dimensions, Nominal D ±.015, -.005	L ±.031
200	.01	.174	.625
	.012	.174	.625
	.015	.174	.625
	.018	.193	.625
	.022	.193	.625
	.027	.235	.625
	.033	.235	.625
	.039	.235	.688
	.047	.235	.688
	.056	.312	.625
	.068	.312	.625
	.082	.312	.688
	.10	.312	.688
	.12	.312	.812
	.15	.312	.812
	.18	.400	.688
	.22	.400	.812
	.27	.400	.812
	.33	.400	.938
	.39	.400	.938
	.47	.400	1.125
	.56	.400	1.312
	.68	.500	1.125
	.82	.500	1.125
	1.0	.562	1.125
	1.2	.562	1.312
	1.5	.562	1.312
	1.8	.562	1.812
	2.0	.562	1.812
	2.2	.562	1.812
	2.7	.670	1.562
	3.0	.750	1.562
	3.3	.750	1.812
	4.0	.750	1.812
	4.7	.750	2.062
	5.0	.750	2.062
	5.6	.750	2.312
	6.0	.750	2.312
	7.0	1.000	1.812
	8.2	1.000	2.062
	10.0	1.000	2.312

Case size of 1" dia for 8.2 mfd and 10 mfd uses #18 AWG leads

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Table I. Capacitor characteristics and dimensions (cont'd)

DC Rated Voltage	Cap Value in µf	Dimensions, Nominal D +.015, -.005	L ±.031
400	.01	.235	.688
	.012	.235	.812
	.015	.235	.812
	.018	.235	.812
	.022	.312	.688
	.027	.312	.812
	.033	.312	.812
	.039	.312	.938
	.047	.400	.812
	.056	.400	.812
	.068	.400	.938
	.082	.400	.938
	.10	.400	1.125
	.12	.400	1.312
	.15	.400	1.312
	.18	.562	1.125
	.22	.562	1.125
	.27	.562	1.312
	.33	.562	1.562
	.39	.562	1.562
	.47	.562	1.812
	.56	.670	1.562
	.68	.670	1.812
	.82	.750	1.812
	1.0	.750	2.062
	1.2	1.000	1.812
	1.5	1.000	1.812
	1.8	1.000	2.062
	2.0	1.000	2.062
	2.2	1.000	2.312
	2.5	1.000	2.562
	2.7	1.000	2.562

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Table I. Capacitor characteristics and dimensions (cont'd)

DC Rated Voltage	Cap Value in µf	Dimensions, Nominal D +.015, -.005	L ±.031
600	.01	.312	.812
	.012	.312	.812
	.015	.400	.812
	.018	.400	.812
	.022	.400	.812
	.027	.400	.937
	.033	.400	.937
	.039	.400	1.125
	.047	.400	1.125
	.056	.400	1.312
	.068	.400	1.312
	.082	.562	1.125
	.10	.562	1.125
	.12	.562	1.312
	.15	.562	1.312
	.18	.562	1.562
	.22	.562	1.812
	.27	.670	1.562
	.33	.670	1.812
	.39	.670	1.812
	.47	.750	1.812
	.56	.750	2.062
	.68	.750	2.312
	.82	1.000	1.812
	1.0	1.000	2.062

1/ Dimensions are bare case sizes (see Figure 1).

2/ Dimensions are for Circuit 1. For Circuit 3, deduct .062" from length.

3/ See Table II for metric equivalents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Table II. Metric equivalents of decimal inches 1/

Inches	MM	Inches	MM	Inches	MM
.005	.13	.670	17.02	1.562	39.67
.015	.38	.688	17.48	1.624	41.25
.031	.79	.750	19.05	1.812	46.02
.175	4.45	.812	20.62	1.874	47.60
.195	4.95	.874	22.20	2.062	52.37
.235	5.97	1.000	25.40	2.124	53.95
.312	7.92	1.062	26.97	2.312	58.72
.400	10.16	1.124	28.55	2.374	60.30
.500	12.70	1.312	33.32	2.562	65.07
.562	14.27	1.374	34.90	2.624	66.65

1/ Metric equivalents (to the nearest .01 millimeter) are given for general information only, and are based upon 1 inch = 25.4 mm.

Table III. Insulation resistance

In Megohms:	
At 25°C ±3°C (need not exceed)	250,000
At 85°C +4, -0°C (need not exceed)	25,000
At 125°C +4, -0°C (need not exceed)	15,000
In Megohms X Microfarads (minimum):	
At 25°C ±3°C	100,000
At 85°C +4, -0°C	6,000
At 125°C +4, -0°C	1,000

Barometric pressure (reduced): Method 105 of MIL-STD-202, Condition D (100,000 feet). 125 percent of rated voltage applied. See MIL-C-39022 for voltage limitations.

Vibration, High frequency: Method 204 of MIL-STD-202, Condition D (20G). 50 percent of rated voltage applied.

Salt spray (corrosion): Method 101 of MIL-STD-202, Condition B (48 hours). Salt solution - 5 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Immersion: Method 104 of MIL-STD-202, Condition C.

DWV:

Insulating sleeves - 4000 VDC, minimum.
Terminal to terminal - 150 percent of rated voltage.
Terminals to case (when case is not a terminal) - 200 percent of rated voltage.

IR:

Insulating sleeves - 100 megohms, minimum.
Terminal to terminal - 60 percent of value specified in Table III.
Terminals to case (when case is not a terminal) - 30,000 megohms, minimum.

CAP: Within ±3 percent of initial value.

DF: Not more than 110 percent of initial limit.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, Condition I.
50 percent of rated voltage applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV, IR, CAP, and DF - Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, Conditions A and D.

Condition A - Applied force 5 pounds
Condition D - 3 rotations of 360 degrees

Low temperature and CAP change with temperature:

Low temperature: -55° +0, -3°C for 48 ±4 hours with rated voltage applied.

CAP change with temperature:

At -55° +0, -3°C : +0, -2 percent
At +85° +4, -0°C: ±1 percent
At +125° +4, -0°C: ±2 percent

Fault count: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

Fungus: Method 508 of MIL-STD-810, Procedure I.

Resistance to soldering heat: Method 210 of MIL-STD-202, Condition B (260 ±5°C for 10 ±1 seconds).

IR: See Table III

CAP: Within ±3 percent of initial value

DF: Not more than initial limit

Life: Method 108 of MIL-STD-202.

Qualification:

Accelerated conditions: 140 percent of rated voltage for 2000 +72, -0 hours at 85°C. End of life requirements as specified for rated conditions.

Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours at 85°C.

DF (at 85°C +4, -0°C) between first 24 and 48 hours of test: Not greater than initial limit.

DF (at 85° +4, -0°C) during last 48 hours of test: 0.8 percent, maximum.

IR:

Insulating sleeves - 100 megohms, minimum

Terminal to terminal - not less than 30,000 megohm-microfarads or 75,000 megohms

Terminals to case (when case is not a terminal) - 30,000 megohms, minimum.

CAP: Within ±5 percent of initial value.

DF (at +25 ±3°C) after Life: 0.5 percent maximum.

Extended Life:

Accelerated conditions: 2,000 +72, -0 hours.

Rated conditions: 10,000 +96, -0 hours.

IR, CAP, and DF - Same requirements as for qualification life test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

125°C verification life test: 85°C rated voltage for 2,000 +72, -0 hours at 125°C. 25 pieces representative of production shall be tested every six months.

IR, CAP, and DF - Same requirements as for qualification Life Test.

Marking: In accordance with MIL-C-39022

Part number: M39022/10 - (Type designation as detailed below)

Explanation of type designation:

M39022/10 - A 104 G M

- M: Failure rate level in percent per 1000 hours: M = 1.0 percent, P = 0.1 percent, R = 0.01 percent, S = 0.001 percent
- G: Capacitance tolerance: K = ±10 percent, J = ±5 percent, G = ±2 percent, F = ±1 percent
- 104: Capacitance: Expressed in picofarads. The first two digits are significant figures; the third is the number of zeros to follow.
- A: Circuit and voltage code (see Table IV).

Table IV. Circuit and Voltage Codes

CODE	CIRCUIT	VOLTAGE
A	1	50
B	3	50
C	1	100
D	3	100
E	1	200
F	3	200
G	1	400
H	3	400
J	1	600
K	3	600

Preparing Activity:
Navy - EC
Project 5910-N264

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command (ELEX 8111)
Washington, DC 20360

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS ... M, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.